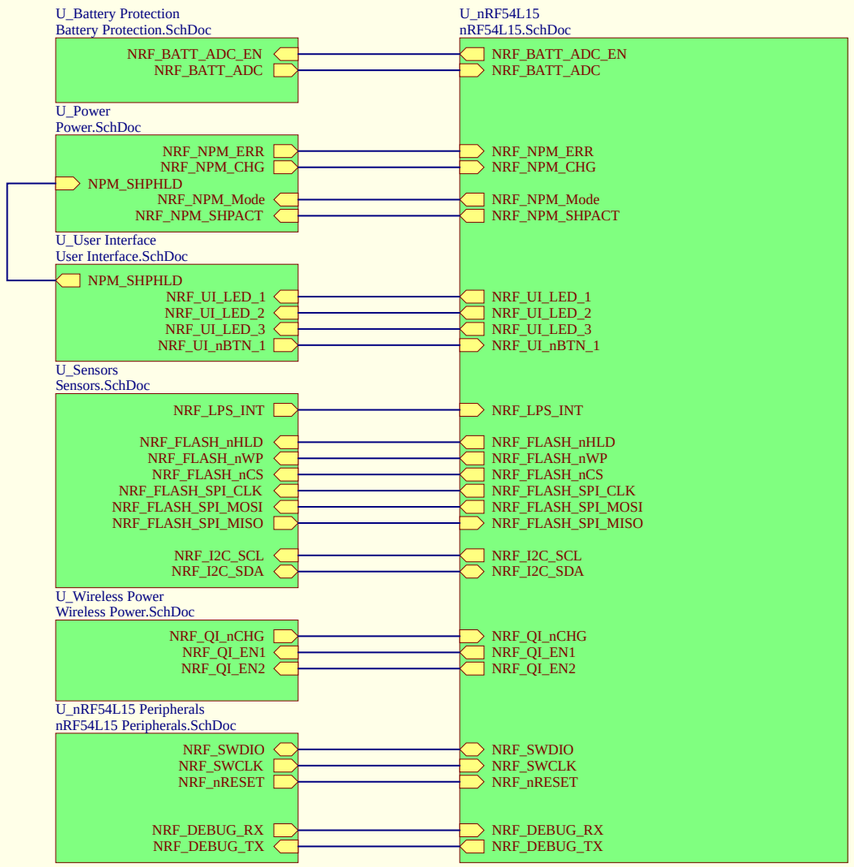


I2C Address Table

I2C0

Device	*Part Number*	*Address (7bit)*	*Power Rail*
Temperature and Relative	SHT40-AD1B-R3	0x44	3V0
CO2 Sensor	STCC4	0x64	3V0
Pressure Sensor	LPS22HHTR	0x5D	3V0

Place fiducials outside board outline on panel

FID1 FID2 FID3

TP7 TP8

0V 0V

M1

1	2	3	4

LAYER CLOCK 4

PWB1



AH-020-0001-PWB Rev3A

U_Revison History A3
Revison History A3.SchDoc

PWA Title:
CO2 Monitor

Genesys Electronics Design
ABN 62 112 253 424
Unit 5, 33 Ryde Rd, Pymble
NSW, 2073, Australia
Ph: +61 2 9496 8900

Sheet Title:
Top

Size: A3	Number: AH-013-0001-SCH	Revision: 3A
Print Date: 1/07/2026 11:58:29 AM		Sheet 1 of 9
File: Top.SchDoc		
Drawn: G.T	Approved: G.S.	Approved Date: 25/06/2026

1S LiPo Protection IC

+VBATT

+VBATT_PROT

TVS on +VABTT_PROT output pin

Fuse: UL File No. E10480

F1

250mA / 32V
0467.250NR
SMT 0603, Height 0.5mm

TP3

Mechanically can not be attached backwards. Thus, no reverse polarity.

TP9

BATT1
V+
V-
182030 75mAh

R1
330R
0402

C1
100n
0402

TP24

U1
BQ29700DSER
BAT
VSS
DOUT
COUT
V-
NC

Over charge detection = 4.350V
Over charge release = 4.150V
Over Discharge detection = 2.3V
Over Discharge relaese = 3.0V
Over current detection voltage = 0.2V

TP4

TP5

TP11

R2
2k2
0402

Q1
PMF63UNE

Q2
PMF63UNE

0V

Rds = 50mohm each
System Imax = 1A
System Iavg = <50uA

MOSFET Vgs(ths) has to be greater than 400mV and less than 4.350V
Our MOSFET Vgs(ths) = 1V @ 25deg (min 500mV)
Vgs(thsmax) = 1V
Vgs(max) = 8V

Voltage Measurement

+VBATT_PROT

R41
1M0
0402

Q4B
DMC2400UV

Q4A
DMC2400UV

NRF_BATT_ADC_EN

NRF_BATT_ADC_EN

R42
1M0
0402

R43
1M0
0402

0V

0V

NRF_BATT_ADC

NRF_BATT_ADC

R44
1M5
0402

0V

Scaling factor of 0.6.
Vbatt 4.2V = Vadc 2.52V
Vbatt 3.7V = Vadc 2.22V
Vbatt 3.3V = Vadc 1.98V

COMMERCIAL-IN-CONFIDENCE

PWA Title:
CO2 Monitor

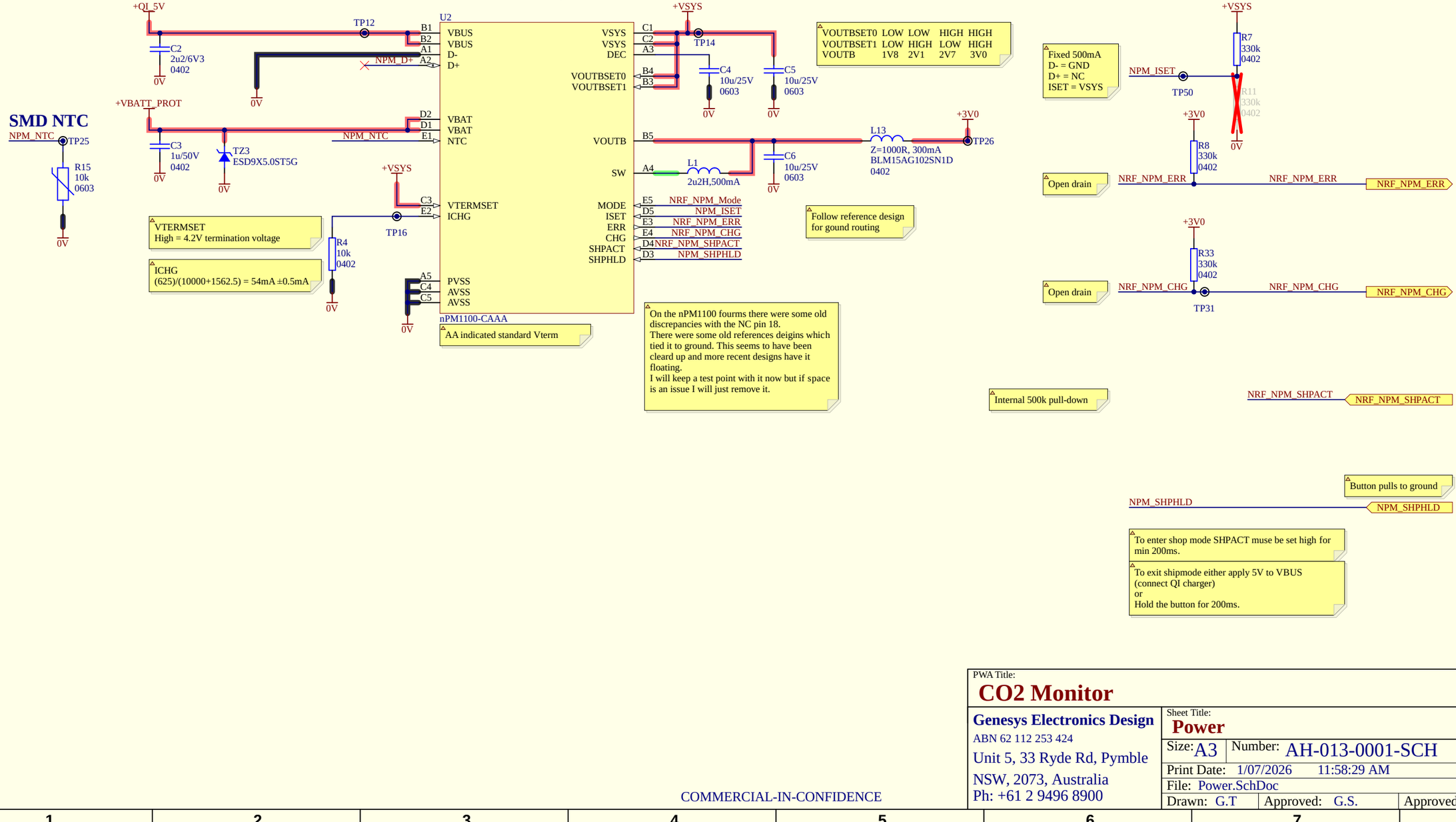
Genesys Electronics Design
ABN 62 112 253 424
Unit 5, 33 Ryde Rd, Pymble
NSW, 2073, Australia
Ph: +61 2 9496 8900

Sheet Title:
Battery Protection

Size: A3	Number: AH-013-0001-SCH	Revision: 3A
Print Date: 1/07/2026 11:58:29 AM		Sheet 2 of 9
File: Battery Protection.SchDoc		
Drawn: G.T	Approved: G.S.	Approved Date: 25/06/2026

COMMERCIAL-IN-CONFIDENCE

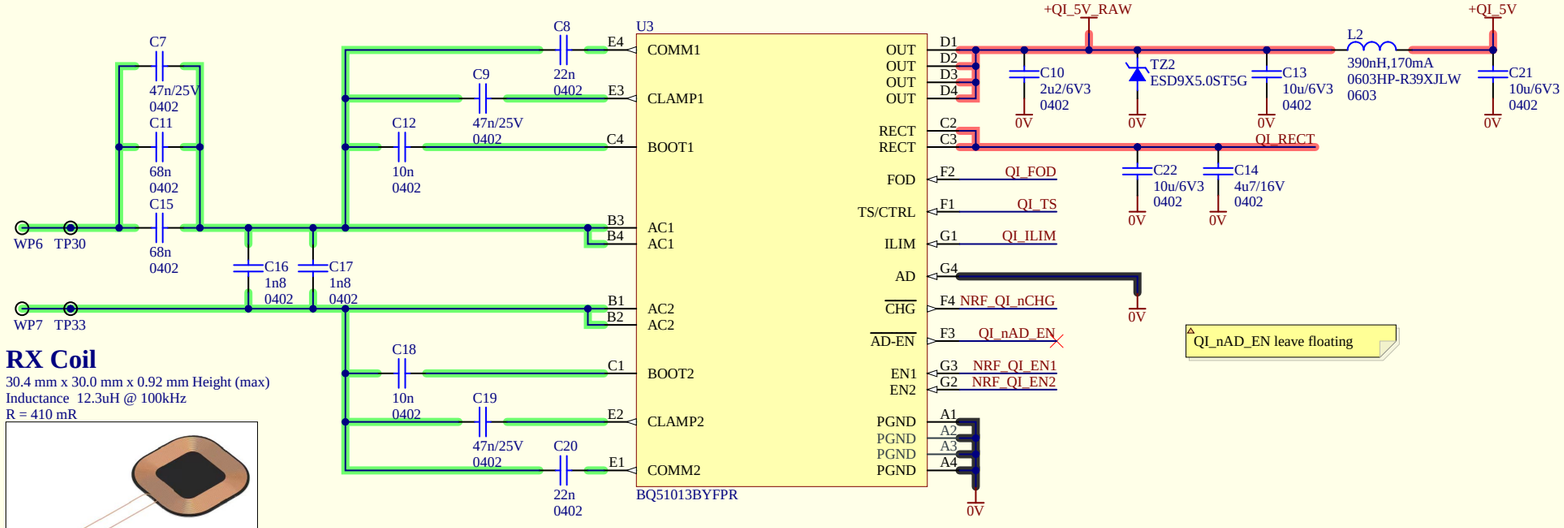
Power Management



PWA Title:	**CO2 Monitor**
Sheet Title:	**Power**
Size: A3	Number: AH-013-0001-SCH
Revision: 3A	
Print Date: 1/07/2026 11:58:29 AM	Sheet 3 of 9
File: Power.SchDoc	
Drawn: G.T	Approved: G.S.
Approved Date: 25/06/2026	

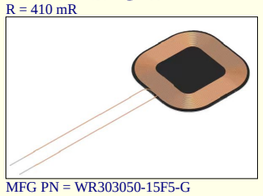
Genesys Electronics Design
ABN 62 112 253 424
Unit 5, 33 Ryde Rd, Pymble
NSW, 2073, Australia
Ph: +61 2 9496 8900

Wireless Power Reciever



RX Coil

30.4 mm x 30.0 mm x 0.92 mm Height (max)
Inductance 12.3uH @ 100kHz
R = 410 mR



MFG PN = WR303050-15F5-G

Place close togeather
80kHz -3dB cutoff

QI_nAD_EN leave floating

SMD NTC

QI_TS TP29
R18 10k 0603
0V

Additional RTC for monitoring board temp while charging. The main battery NTC lives on the power sheet

QI_RECT TP52
R13 20k / 0.1% 0402
QI_ILIM TP53
R14 2k7 0402
QI_FOD TP54
R16 100R / 0.1% 0402
R17 91R 0402
0V

Adds option for tuning FOD
Currently using default values (tesed with default values tested with devkit)

ILIM
K=314
RLIM=314/ILIM

Iout(A)	R(ohm)
0.62	510
0.56	560
0.51	620
0.42	750
0.35	910
0.31	1,000
0.24	1,300
0.17	1,800
0.12	2,700
0.06	5,100

TP35 NRF_QI_EN1 NRF_QI_EN1
TP36 NRF_QI_EN2 NRF_QI_EN2

Internal 200k pull-down

0 0 = Enabled with handshake

+3V0
R19 330k 0402
5mA capable open-drain
NRF_QI_nCHG NRF_QI_nCHG NRF_QI_nCHG

Genesys Electronics Design
ABN 62 112 253 424
Unit 5, 33 Ryde Rd, Pymble
NSW, 2073, Australia
Ph: +61 2 9496 8900

PWA Title: **CO2 Monitor**

Sheet Title: **Wireless Power**

Size: A3	Number: AH-013-0001-SCH	Revision: 3A
Print Date: 1/07/2026 11:58:29 AM		Sheet 4 of 9
File: Wireless Power.SchDoc		
Drawn: G.T	Approved: G.S.	Approved Date: 25/06/2026

NRF_BATT_ADC_EN
NRF_BATT_ADC
NRF_NPM_Mode
NRF_NPM_ERR
NRF_NPM_CHG
NRF_NPM_SHPACT
NRF_QI_EN1
NRF_QI_EN2
NRF_QI_nCHG
NRF_LPS_INT
NRF_UI_LED_1
NRF_UI_LED_2
NRF_UI_LED_3
NRF_UI_nBTN_1
NRF_FLASH_nHLD
NRF_FLASH_nWP
NRF_FLASH_nCS
NRF_FLASH_SPI_CLK
NRF_FLASH_SPI_MOSI
NRF_FLASH_SPI_MISO
NRF_I2C_SCL
NRF_I2C_SDA
NRF_SWDIO
NRF_SWCLK
NRF_DEBUG_TX
NRF_DEBUG_RX
NRF_nRESET

nRF54 Pins

32.768kHz XAL
Can use external or configurable internal capacitors
Tested with development kit

X2 32.768kHz

U7B

NRF_QI_nCHG	TP18	23	P0.00 (Blue Pins Are Dedicated Clock Pins)
NRF_NPM_ERR	TP19	24	P0.01
NRF_UI_nBTN_1	TP20	27	P0.02
NRF_I2C_SCL	TP21	28	P0.03/GRTC_PWM
NRF_I2C_SDA	TP22	29	P0.04/GRTC_CLKOUT32K
NRF_XAL_1		1	P1.00/XL1
NRF_XAL_2		2	P1.01/XL2
NRF_DEBUG_TX		3	P1.02/NFC1
NRF_DEBUG_RX		4	P.1.03/NFC2
NRF_BATT_ADC_EN	TP27	5	P1.04/ASO0/AIN0
NRF_BATT_ADC	TP28	6	P1.05/ASI0/RADIO6/AIN1
NRF_FLASH_nHLD		7	P1.06/ASO0/AIN2
NRF_FLASH_nWP		8	P1.07/ASI1/AIN3
NRF_NPM_CHG		9	P1.08/CLKOUTFAST/EXTREF
NRF_NPM_Mode	TP32	37	P1.09/ASO2/RADIO0
		38	P1.10/ASI2/RADIO1
		39	P1.11/ASO3/RADIO2/AIN4
NRF_QI_EN1		40	P1.12/ASI3/RADIO3/AIN5
NRF_QI_EN2		41	P1.13/RADIO4/AIN6
		42	P1.14/RADIO5/AIN7
		11	P2.00/SPIM_DCX/UARTE_RXD/FLPR.4/QSPI_D3
NRF_SPI_CLK	TP2	12	P2.01/SPIM_SCK/SPIS_SCK/FLPR.0/QSPI_SCK
NRF_SPI_MOSI		13	P2.02/SPIM_SDO/SPIS_SDO/UARTE_TXD/FLPR.1/QSPI_D0
NRF_NPM_SHPACT	TP41	14	P2.03/FLPR.3/QSPI_D2
NRF_FLASH_SPI_MISO	TP42	15	P2.04/SPIM_SDI/SPIS_SDI/UARTE_CTS/FLPR.2/QSPI_D1
NRF_IC_FLASH_nCS		16	P2.05/SPIM_CS/SPIS_CS/UARTE_RTS/FLPR.5/QSPI_CS
NRF_UI_LED_2	TP44	17	P2.06/FLPR.6/SPIM_SCK/SPIS_SCK/TRACECLK
		18	P2.07/FLPR.7/TRACEDATA[0]/SWO/SPIM_DCX/UARTE_RXD
NRF_UI_LED_1	TP46	19	P2.08/FLPR.8/TRACEDATA[1]/SPIM_SDO/SPIS_SDO/UARTE_TXD
NRF_UI_LED_3	TP47	20	P2.09/FLPR.9/TRACEDATA[2]/SPIM_SDI/SPIS_SDI/UIARTE_CTS
NRF_LPS_INT	TP48	21	P2.10/FLPR.10/TRACEDATA[3]/SPIM_CS/SPIS_CS/UARTE_RTS

nRF54L15-QFAA

NRF_FLASH_SPI_CLK — R45 22R — NRF_SPI_CLK
NRF_FLASH_SPI_MOSI — R46 22R — NRF_SPI_MOSI
NRF_FLASH_nCS TP1 — R47 22R — NRF_IC_FLASH_nCS

nRF54 Low Power Reccomendations

- MCU pins to be actively set to analog mode when not in use.
- Peripherials to be disabled when not in use.
- DCDC regulator to be used.
- Extenral crystal to be used.
- TX power of -8dBm reccomended
- Port 1 should be disabled most of the time
- X2 to power GRTC
- SAADC on as little as possible

nRF54 Power + Misc

+3V0 — L6 Z=1000R, 300mA BLM15AG102SN1D — +nRF_VDD

U7A

VDD (47, 48, 10, 22, 36)
C26 10u/25V, C27 100n, C28 100n, C29 100n, C30 100n

DCC (46) — L3 4u7H, 120mA MLZ1608M4R7WT000 — DECD (45) — C31 2u2/6V3
L4 Z=120R, 200mA BLM18AG121SN1D — DECA (43, 33) — DECRF
C32 2u2/6V3, C33 10n, C34 2n2

VSS (44, 49)

XC1 (34) — X1 32.0MHz — XC2 (35)

32MHz XAL
Can use external or configurable internal capacitors

ANT (31) — NRF_BLE_ANTENNA
VSS_PA (32) — To Pin49

SWDCLK (26) NRF_SWCLK
SWDIO (25) NRF_SWDIO
nRESET (30) — R29 1k0 — NRF_nRESET
C36 3n9

SWCLK: Internal pull-up
SWDIO: Internal pull-down
nRESET: Internal pull-up

nRF54L15-QFAA

BLE Antenna

50R — L8 2n7H,+/-0.1nH — L9 3n6H,+/-0.2nH LQW15AN3N6C10D — L10 3n6H,+/-0.2nH LQW15AN3N6C10D
C38 1p5,+/-0.1p, C39 2p0,+/-0.1p, C40 0p3,+/-0.05p

To Pin32 ONLY
ONLY ground at bottom layer

U9 B39242B9413K610 (IN, GND, GND, OUT, GND)

Chip Antenna Matching Network
R26 0R, R39 0R, C35 3p9,+/-0.25pF, L5 2n7H,+/-0.1nH

CHIP_ANTENNA

Chip Antenna

ANT1 (GND, GND) — ANT3216A063R2400A

Important note on C38:
- Grounding of C38 capacitor must ONLY be connected to Pin32 (VSS_PA) in top layer.
- Pin32 must ONLY be connected to Pin49 (VSS center pad) underneath the IC package

Important note on C39:
- Grounding of C39 capacitor MUST be isolated from ALL ground layers except the bottom ground layer.

PWA Title: **CO2 Monitor**

Genesys Electronics Design
ABN 62 112 253 424
Unit 5, 33 Ryde Rd, Pymble
NSW, 2073, Australia
Ph: +61 2 9496 8900

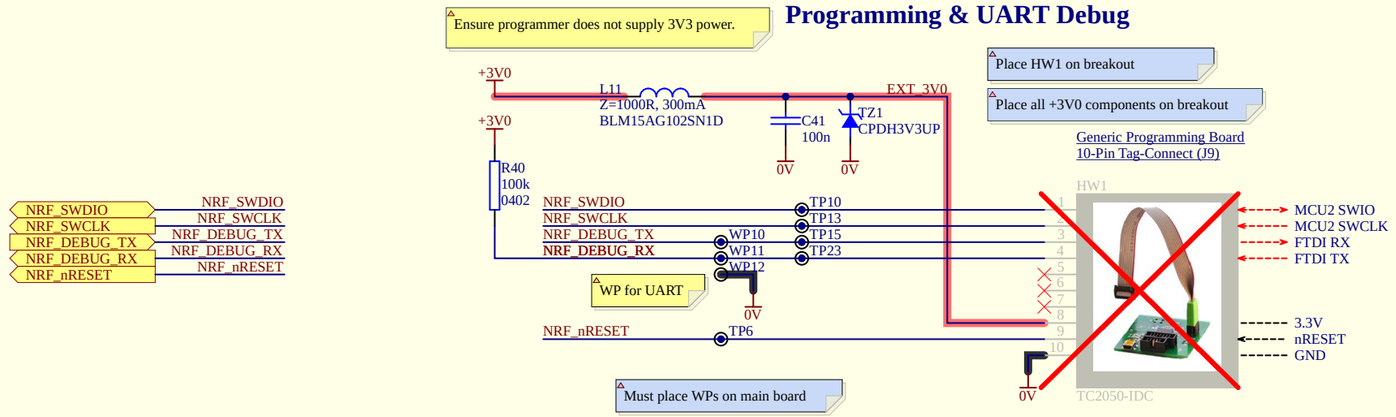
Sheet Title: **nRF54L15**

Size: A3	Number: AH-013-0001-SCH		Revision: 3A
Print Date: 1/07/2026 11:58:29 AM			Sheet 5 of 9
File: nRF54L15.SchDoc			
Drawn: G.T	Approved: G.S.	Approved Date: 25/06/2026	

COMMERCIAL-IN-CONFIDENCE

COMMERCIAL-IN-CONFIDENCE

Programming & UART Debug



PWA Title:	**CO2 Monitor**
Genesys Electronics Design ABN 62 112 253 424 Unit 5, 33 Ryde Rd, Pymble NSW, 2073, Australia Ph: +61 2 9496 8900	Sheet Title: **nRF54L15 Peripherals**
Size: A3 / Number: AH-013-0001-SCH / Revision: 3A	
Print Date: 1/07/2026 11:58:29 AM	Sheet 6 of 9
File: nRF54L15 Peripherals.SchDoc	
Drawn: G.T / Approved: G.S.	Approved Date: 25/06/2026

Temp & Humidity Sensor

I2C Address 0x44 (7bit)

Vcc = 1.08 V to 3.6 V
Operating Current = 2.2 uA (avg)
Idle Current = 1 uA
Temp. Range = -40C to 125C

CO2 Sensor

I2C Address 0x64 (7bit)

Vcc = 2.7 V to 5.5 V
Operating Current = 500 uA (avg 10s)
Sleep Current = 1 uA
Temp. Range = 0C to 40C

Pressure Sensor

I2C Address 0x5D (7bit)

Vcc = 2.7 V to 5.5 V
Operating Current = 4 uA (avg)
Sleep Current = 0.9 uA
Temp. Range = -40C to 85C

Set to push-pull in firmware

Serial FLASH Memory

FLASH used for storing information required by the unit's software, including graphics images and logged data.

COMMERCIAL-IN-CONFIDENCE

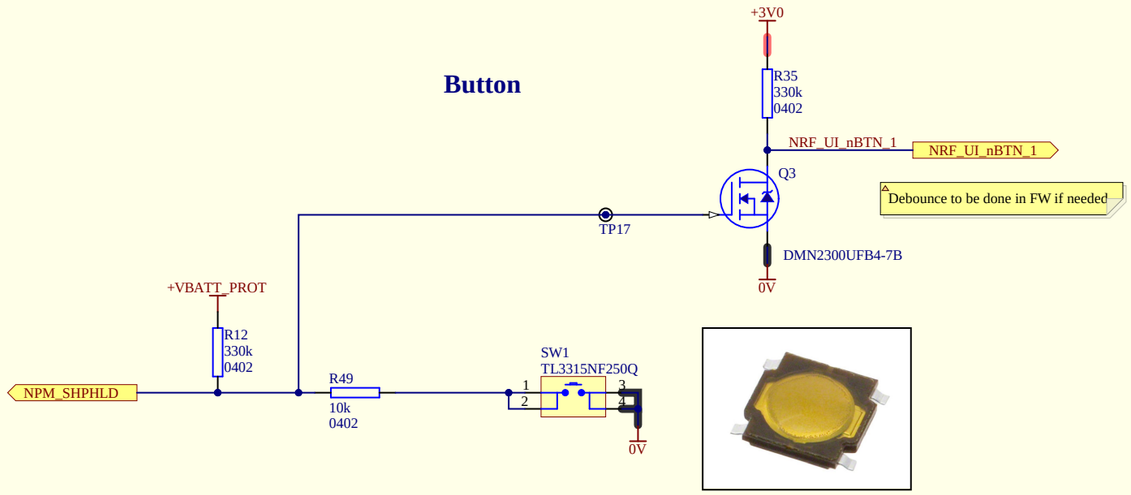
PWA Title: **CO2 Monitor**

Genesys Electronics Design
ABN 62 112 253 424
Unit 5, 33 Ryde Rd, Pymble
NSW, 2073, Australia
Ph: +61 2 9496 8900

Sheet Title: **Sensors**

Size: A3	Number: AH-013-0001-SCH	Revision: 3A
Print Date: 1/07/2026 11:58:29 AM		Sheet 7 of 9
File: Sensors.SchDoc		
Drawn: G.T	Approved: G.S.	Approved Date: 25/06/2026

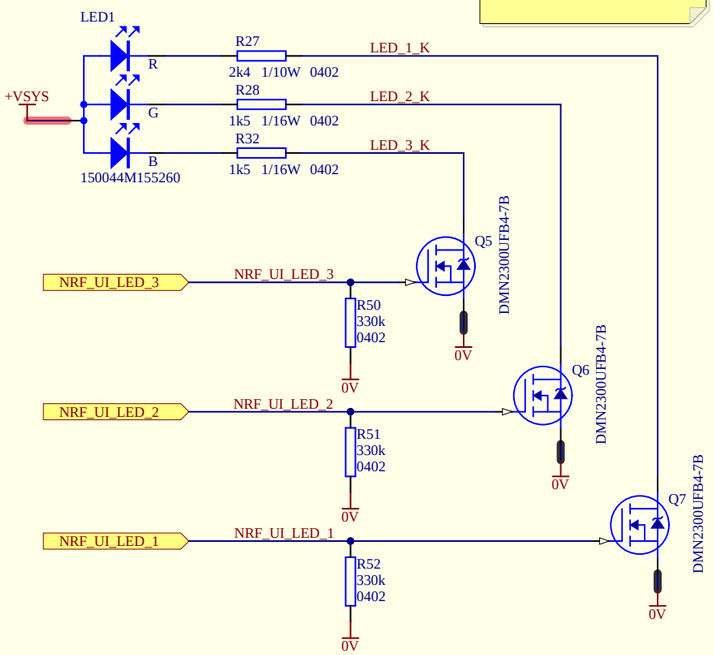
Button



LED Indicator

LED Drive Current = 5 mA (typ)
LED Pin Voltage = 0.5 V to VSYS

CL-505S-X-SD-T:
Ired = Vf = 1.85, 1mA @4V2
Igreen = Vf = 2.70, 1mA @4V2
Iblue = Vf = 2.75, 1mA @4V2



PWA Title:
CO2 Monitor

Genesys Electronics Design
ABN 62 112 253 424
Unit 5, 33 Ryde Rd, Pymble
NSW, 2073, Australia
Ph: +61 2 9496 8900

Sheet Title: **User Interface**

Size: A3	Number: AH-013-0001-SCH	Revision: 3A
Print Date: 1/07/2026 11:58:29 AM		Sheet 8 of 9
File: User Interface.SchDoc		
Drawn: G.T	Approved: G.S.	Approved Date: 25/06/2026

REV	CHANGES	DATE	ENG	APP
1A	Initial Release	29-Oct-25	GT	GS
2A	Changed TZ2 & TZ3 from SMB6F5.0AY to ESD9X5.0ST5G	20-Nov-25	GT	
3A	Sheet 2: 1. Removed R3 and J1 VBATT testing jumper. Sheet 3: 1. Changed U2 to smaller BGA package. 2. Removed R48, R5, R9, R34 and R10 0R jumpers. 3. Removed J2 +3V0 testing jumper. Sheet 4: 1. Change U3 to smaller BGA package. 2. Removed R20, R30 and R31 0R jumpers. Sheet 5: 1. SWO signal from backup programming header no longer routed to nRF54. 2. Swapped NRF_UI_nBTN_1 and NRF_LPS_INT nRF54 pinout. Sheet 6: 1. J3 removed Sheet 7: 1. R38, L7, R53, L13 removed 2. Changed R36 designator to R5 3. R22 designator changed to R3 and R23 designator changed to R10 Sheet 8: 1. TZ4 removed 2. LED1 changed to PN 150044M155260	25-Jun-26	WC	GS

REV	CHANGES	DATE	ENG	APP

COMMERCIAL-IN-CONFIDENCE

PWA Title:
Standard Build

Genesys Electronics Design
ABN 62 112 253 424
Unit 5, 33 Ryde Rd, Pymble
NSW, 2073, Australia
Ph: +61 2 9496 8900

Sheet Title:
Revision History

Size: A3	Master: AH-013-0001-SCH	Revision: 3A
Print Date: 1/07/2026 11:58:29 AM		Sheet 9 of 9
File: Revison History A3.SchDoc		
Drawn: G.T	Approved: G.S.	Approved Date: 25/06/2026